
RoboSynChallenge: Mastering Real-World Dexterity via Generalizing Synthesized Manipulation Skills

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Abstract

Achieving generalizable robotic manipulation remains a central challenge in embodied intelligence. Despite rapid advances in model architectures and learning algorithms, progress is often limited by the scarcity and narrow diversity of real-world data. The **RoboSynChallenge** competition introduces a unified benchmark to evaluate and advance the *generalizability* of manipulation policies across a spectrum of tasks, environments, and difficulty levels. To alleviate the shortage of realistic data, the challenge integrates large-scale synthetic data generation with standardized real-world robotic evaluation. Participants are encouraged to leverage synthesized state-action trials to improve general-purpose policy learning, while final assessments are conducted exclusively on unseen real-world manipulation environments. Baseline implementations, including Transformer-, Diffusion-, Vision-Language-Action, and World-Action-Model-based policies, are provided to ensure reproducibility and comparability. By coupling scalable simulation-based training with rigorous real-world validation, **RoboSynChallenge** aims to foster the development of broadly capable, data-efficient, and adaptable manipulation systems, thereby paving the way toward truly general robotic intelligence.

Keywords Embodied AI, Sim2Real Transfer, Synthetic Data, Dexterous Manipulation

1 Competition description

Background. Developing scalable and precise robotic manipulation policies represents a critical step toward realizing the vision of *Embodied Artificial Intelligence (EAI)* [1]. As a technology with broad societal and industrial impact, scalable robotic manipulation holds the potential to reduce operational costs and enhance productivity across manufacturing, household, and healthcare domains [2].

Data-driven learning methods have emerged as a principled approach for developing general-purpose robotic agents that can operate flexibly across diverse tasks and environments. In recent years, *generalist policy models* have been advanced through the use of Transformers [3], diffusion models [4], Vision-Language-Action (VLA) frameworks [5], and World Action Models (WAMs) [6, 7, 8, 9]. These architectures enable end-to-end learning pipelines that map multimodal observations and language instructions directly to continuous robot actions [10], paving the way for more autonomous, instruction-following control systems.

As research progresses toward developing generalist manipulation policies, a central question arises: how can we fairly and comprehensively evaluate and benchmark such general-purpose robotic systems? Existing benchmarks are typically categorized as either simulation-based [11, 12, 13, 14] or real-world [15, 16, 17, 18]. However, unlike vision or language models, which can draw upon vast internet-scale datasets, real-world robotic benchmarks rely on data obtained through physical

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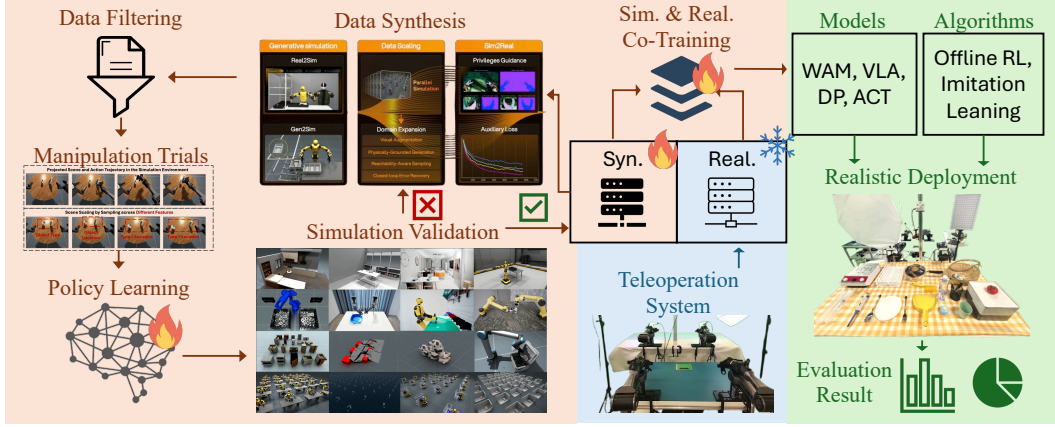


Figure 1: The pipeline of RoboSynChallenge, which provides a comprehensive data generation framework for synthesizing manipulation trials within a simulated environment (in orange background), thereby expanding the training datasets. These synthetic data, together with a smaller amount of static real-world data collected through teleoperation (in blue background), can be used for simulation-to-real co-training. This approach helps mitigate the limitations of scarce real data when deploying trained manipulation policies in the real world (in green background).

interaction, making data collection costly, slow, and tied to specific hardware setups [5]. While simulation facilitates scalable data generation [19, 20, 21, 22], mismatches in dynamics, kinematics, and sensing inevitably yield a simulation-to-reality (Sim2Real) gap, resulting in degraded real-world performance [14, 23, 24]. Bridging this Sim2Real gap thus remains a fundamental challenge in embodied intelligence research. Moreover, most existing benchmarks rely on fixed datasets, which assume static and controlled environments. This design misaligns with the objective of building generalist policies that must generalize to unseen objects and diverse environments.

Impact. In this proposal, we push the frontier a step further by introducing RoboSynChallenge, a competition designed to quantify *how effectively simulated data can improve the generalizability of manipulation policies when real-world data is scarce*. To achieve this goal, RoboSynChallenge provides a comprehensive data-generation and evaluation pipeline that integrates scalable simulation environments with real-world physical setups for performance validation. The simulation platform enables practitioners to flexibly configure and scale dataset generation across diverse conditions, while the real-world evaluation protocol encompasses multiple levels of difficulty. Together, these components facilitate a rigorous and systematic analysis of Sim2Real transferability.

The RoboSynChallenge is envisioned as an open and scalable benchmark to advance research on generalizable robot learning across simulated and real-world domains. The anticipated impact of RoboSynChallenge is multifaceted as follows:

- 1) **Standardized Benchmark.** RoboSynChallenge seeks to establish a standardized real-world benchmark that spans the entire dexterous manipulation policy pipeline, from large-scale data generation and policy learning with synthesized data, to rigorous evaluation under physical environments.
- 2) **Real-world Reproducibility.** RoboSynChallenge strengthens the reproducibility of robotic control policies through standardized data synthesis protocols and evaluation frameworks, ensuring consistent benchmarking and fair comparison across real-world environments.
- 3) **Opensource Tools and Baselines.** RoboSynChallenge provides publicly available baseline models, data synthesis, and training pipelines, fostering transparent evaluation, reproducible experimentation, and collaborative progress within the robotics research community.
- 4) **Toward Generalizable Manipulation.** RoboSynChallenge provides a foundational platform toward generalizable manipulation policies by coupling large-scale synthetic data generation with standardized real-world evaluation, enabling systematic study of scalable policy generalization.

1.1 Novelty

As a new competition, RoboSynChallenge introduces substantial differences compared to previous and ongoing challenges. Our key innovations are as follows:

1) Benchmarking Sim2Real Transferability. RoboSynChallenge aims to fill a critical gap in the field by providing the first standardized benchmark for Sim2Real transferability. It systematically evaluates how well simulated data and policies support scaling to complex, real-world environments. Beyond measuring raw performance, RoboSynChallenge assesses robustness, adaptability, and generalization under real-world noise and domain shifts. This approach fundamentally differs from prior Sim2Sim or Real2Real benchmarks that operate within a single modality. It represents a significant step toward unified Sim2Real evaluation and understanding the limits of simulation-based learning.

2) Generative Data Streaming. Unlike prior benchmarks that rely on a fixed, human-crafted dataset, RoboSynChallenge integrates an automated data generation pipeline [25]. The system procedurally synthesizes simulation environments and automatically generates motion trails in a closed loop. Moreover, to bridge the Sim2Real gap, practitioners can use the provided realistic data as input to style transfer models [26], video generation models [27], or scene augmentation techniques [28, 19] to synthesize diverse motion trajectories with high visual fidelity. This enables large-scale, diverse, and efficient data acquisition without manual curation.

3) Real-World Evaluation. RoboSynChallenge integrates real-world robotic evaluation as part of its official leaderboard. Submitted policies are deployed and tested on standardized physical robot platforms using the same task definitions and evaluation metrics as in simulation. This unified testbed directly measures Sim2Real performance and provides a transparent benchmark for real-world robustness, enabling participants to assess both efficiency and adaptability under tangible constraints.

Table 1 highlights the distinctions between our approach and existing benchmarks. Unlike previous benchmarks that were limited to single-arm settings, simulations, or real-world datasets alone, RoboSynChallenge unifies bimanual manipulation across both simulated and realistic environments. Crucially, it leverages simulated data in a generative manner to augment real-world interactions, enabling richer policy learning and improved generalization. This hybrid paradigm bridges the data efficiency of simulation with the robustness of real-world performance, extending the scope of dexterous manipulation beyond existing competitive and benchmark platforms.

Table 1: The comparison to other competition and benchmarks for dexterous manipulation.

Competition	Rigid Objects	Articulated Objects	Assembling Objects	Tool Using	Manip. Setting	Dataset	Training Env.	Evaluation Env.
RLBench [11]	✓	✓	✗	✗	Single-Arm	Offline	Simulated	Simulated
CALVIN [12]	✓	✓	✗	✗	Single-Arm	Offline	Simulated	Simulated
LIBERO [13]	✓	✓	✗	✗	Single-Arm	Offline	Simulated	Simulated
RoboTwin [29]	✓	✓	✗	✓	Both	Offline	Simulated	Simulated
WBCD ³	✓	✓	✓	✗	Bimanual	None	Realistic	Realistic
RoboChallenge [15]	✓	✓	✓	✓	Both	Offline	Realistic	Realistic
RobotArena ∞ [16]	✓	✓	✗	✗	Single-Ar	None	Realistic	Realistic
RobotArena [17]	✓	✓	✗	✓	Single-Arm	Offline	Realistic	Simulated
ManipulationNet [18]	✓	✗	✓	✗	Single-Arm	Offline	Realistic	Realistic
ManipArena ⁴	✓	✓	✗	✗	Bimanual	Offline	Realistic	Realistic
RoboSynChallenge	✓	✓	✓	✓	Bimanual	Generative	Sim. & Real.	Realistic

1.2 Data

The dataset employed in RoboSynChallenge is primarily generated specifically for this competition using the *EmbodiChain* [25] generative simulation framework. It comprises large-scale embodied interaction trajectories, multi-modal sensory streams (including RGB-D data, proprioceptive signals, and physics-based scene metadata), and structured annotations detailing task semantics, contact events, and success outcomes. To moderate the difficulty of Sim2Real transfer, RoboSynChallenge also includes a smaller subset of data collected through teleoperation to provide real-world correspondence for a limited set of scenarios. The detailed generation process is described as follows.

All robot data were produced within robot manipulation environments using a simulation environment, generative methods, and teleoperation. No personally identifiable or human data are included.

Data Collection Protocol. To facilitate the generalizability of manipulation policy in the RoboSynChallenge, we designed protocols for collecting both simulated and real-world datasets.

1) *Real-World Data Collection Protocol.* To construct the real-world dataset, each manipulation task was conducted under five distinct experimental conditions, each further evaluated across four positional variations and three orientation settings, resulting in a total of 60 samples per task. The experimental conditions are designed to introduce variability in background texture, illumination, and

scene complexity, thereby enabling a rigorous assessment of generalization in real-world manipulation scenarios. The detailed configuration of the experimental conditions is summarized in Table 2. Each condition was systematically combined with four positional variations and three orientation settings, ensuring comprehensive coverage of spatial and visual factors across all collected samples.

Table 2: Summary of real-world data collection conditions.

Background	Lighting Setting	Additional	Description
White	Fixed lighting	None	Standard setting with neutral background and lighting.
White	Enhanced lighting	None	Increased illumination to vary lighting conditions.
White	Fixed lighting	2–3 distractors	Includes additional objects to test robustness.
Blue	Fixed lighting	None	Alternative background color to introduce visual contrast.
Yellow	Fixed lighting	None	Textured background used to assess visual generalization.

2) *Synthesized Data Collection Protocol.* RoboSynChallenge integrates Embodichain [25] to scale up the simulated state-action trails. Appendix A.1 shows the details of data generation. Each simulation instance involved systematic perturbations in lighting, object attributes, table properties, background color, and robot configuration. In addition, both intrinsic and extrinsic camera parameters were randomized to simulate different viewpoints and imaging conditions. The randomization schema is summarized in Table 3 in the Appendix. This procedure introduces controlled, multi-level variability, including lighting, materials, geometry, and sensing, which ensures the simulated dataset captures diverse visual and physical conditions for robust policy training and evaluation. We randomly sampled variations to generate 1,000 manipulation trials for each task. The complete data collection pipeline has been released as open-source, *allowing practitioners to flexibly modify and extend the framework according to their specific research needs.*

In addition, since we provide realistic data, practitioners can leverage it to guide video generation models [27] through in-context learning or fine-tuning. The generated data can be further augmented or refined using scene augmentation techniques (e.g., [28, 19]) applied in either simulated or real-world settings. RoboSynChallenge offers flexible integration with such pipelines, facilitating large-scale synthesized data collection.

Confidential Ground Truth and Leakage Prevention. Our evaluation protocol measures the model’s real-world performance using diverse metrics (Section 1.4). This evaluation does not require access to the testing data or any ground-truth labels for verification. Instead, it assesses outcomes directly in the real world. Moreover, the training data consist only of successful trajectories collected in the learning environment. To quantify generalizability, the evaluation employs a held-out testing protocol in which the test environments are out-of-distribution relative to the training observations, ensuring minimal overlap and reducing the risk of dataset leakage.

All training and evaluation datasets will be made freely available to registered competition participants through online platforms (e.g., Hugging Face). The dataset is released under a **Creative Commons Attribution–NonCommercial–ShareAlike 4.0 (CC BY-NC-SA 4.0)** license, permitting research use while preventing commercial redistribution. Each release will include *metadata, documentation, and generation code* (for simulated data), in accordance with the NeurIPS Code of Ethics.

1.3 Tasks and application scenarios

Hardware. As Figure 2 shows, the evaluation of manipulation policy relies on a dual-arm platform built from two AgileX PiPER-X manipulators. Each PiPER-X is a 6-DoF research arm featuring an agile wrist and open interfaces, designed for embodied manipulation and rapid system integration. This platform has been widely adopted in both prior competitions (e.g., RoboTwin [14], WBCD 2025⁵, and WBCD 2026⁶) and recent research works [7, 30, 22, 31].

Manipulation tasks. Under the above hardware system, our design tasks are diverse and involve handling rigid objects, articulated objects, deformable objects, and tools. All tasks are performed based solely on visual observations and the robot’s proprioceptive feedback. An overview of the real-world setup is shown in Figure 2. The workspace consists of an adjustable-height table with replaceable tablecloth textures and configurable lighting. We categorize the tasks as follows:

⁵<https://wbcdcompetition.github.io/2025/index.html>

⁶<https://wbcdcompetition.github.io>

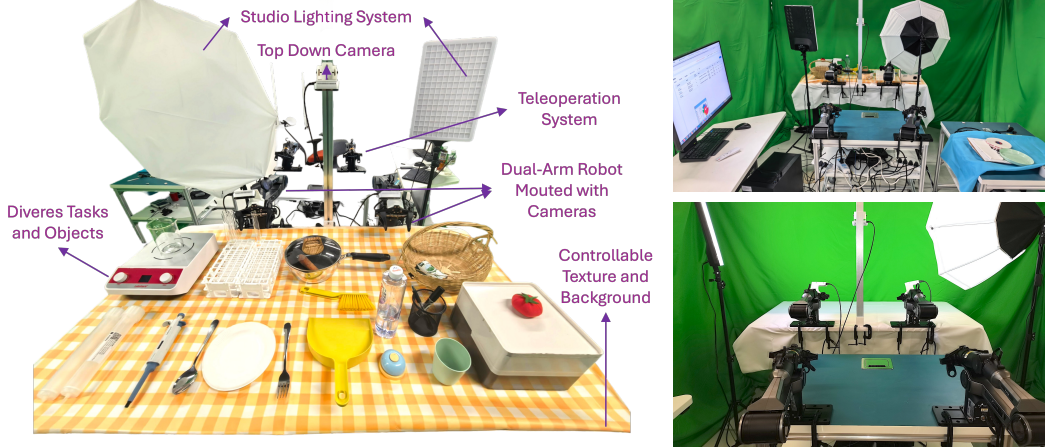


Figure 2: Real-world manipulation setup: The physical platform consists of a dual-arm robot, cameras, lighting, and various objects, enabling evaluation of all tasks included in the benchmark. To facilitate multiple rounds of real-world testing in a reliable and efficient manner, we have prepared *three sets of backup workstations*, each configured identically to the primary setup (see Appendix xxx).

1) *Entry-level tasks* primarily include short-horizon, low-contact-complexity routines with clear affordances. As shown in Figure 3 (green background) and Table 4, they cover table rearrangement, click-bell, water pouring, and handle basket. These tasks emphasize stable perception, reliable grasp execution, and consistent motion trajectories under minimal environmental uncertainty. The focus lies in ensuring repeatable performance of simple action primitives rather than complex planning or intricate contact manipulation.

2) *Mid-level tasks* consist of compound, sequential interactions requiring moderate coordination, adaptive control, and spatial reasoning. As described in Figure 3 (blue background) and Table 4, these include items hand-over, drawer open-and-place, and mixer operating. In contrast to entry-level routines, mid-level tasks introduce variable contact conditions, multi-object dependencies, and partial planning horizons. Evaluation focuses on assessing smooth transition across motion phases, consistent force regulation, and robustness to mild external perturbations.

3) *High-level tasks* comprise fine-grained manipulation and domain-specific operations demanding high precision, multi-stage sequencing, and dynamic adaptation. As outlined in Figure 3 (red background) and Table 4, they include item assembly, pipette manipulation, and sample loading. These tasks require complex reasoning over ordered procedural steps, precise end-effector control, and environmental feedback integration. Benchmarking evaluates success through accuracy of placement or alignment, procedural consistency, and recovery capability under minor execution deviations.

Across these three levels, RoboSynChallenge provides in-distribution testing (with respect to real-world training data) for model development and out-of-distribution testing, acting as a held-out benchmark to evaluate generalization quality throughout the competition.

1.4 Metrics

In our experiments, we present the evaluation protocol and metrics as follows.

Evaluation Protocol. We evaluate dexterous manipulation performance across multiple tasks under the following real-world variations: 1) *Background variation*: three table textures (wood, blue fabric, yellow grid). 2) *Lighting variation*: three distinct light positions with varying illumination colors. 3) *Object variation*: both *seen* and *unseen* object instances within the same task category, evaluating generalization to novel appearances. 4) *Distractor presence*: task-irrelevant objects placed in the scene at three difficulty levels (2, 4, and 8 distractors), sampled from a separate object set distinct from task objects. 5) *Spatial generalization*: evaluation on unseen positions within a predefined 3×3 grid. For each task, we first evaluate the policy under a canonical base configuration. We then systematically vary one factor at a time while keeping all other factors fixed. For each configuration, we evaluate the task under multiple object positions by applying small random shifts, and report the resulting success rate across these variations.

Evaluation Metrics. Based on the aforementioned evaluation protocol, we evaluate the performance of a dexterous policy according to the following methods: 1) *Success Rate (SR)* (in percentage)

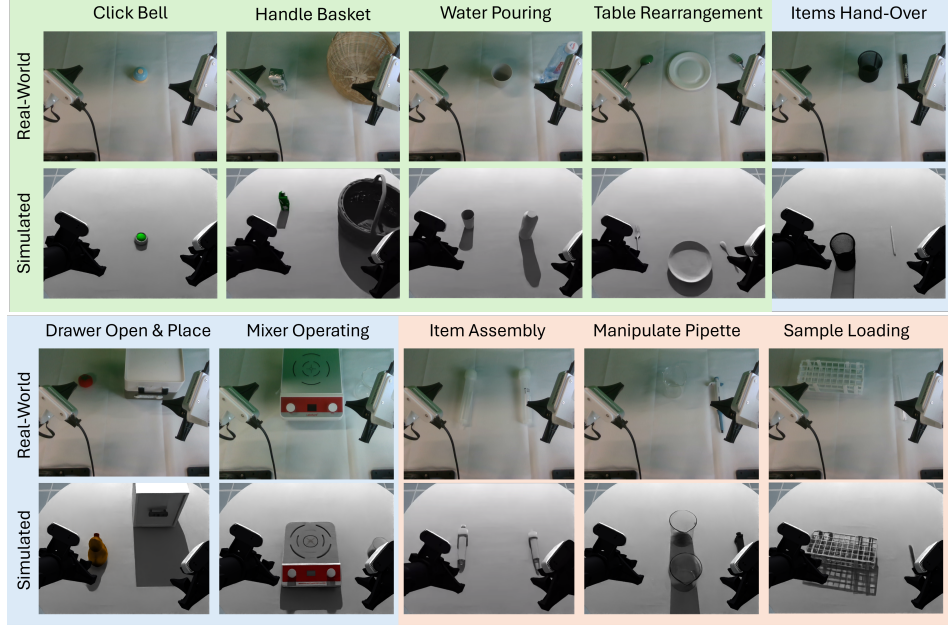


Figure 3: Visualizations of the realistic environments (upper row) and their corresponding simulated environments (lower row) across the three task categories: entry-level (green background), mid-level (blue background), and high-level (red background).

measures how often a dexterous manipulation policy completes a task according to predefined success criteria (see Table 4). 2) *Inference Time* measures how quickly the policy produces actionable commands during inference; lighter models typically run faster. All models are deployed on the same machine with an NVIDIA RTX 4090 GPU to ensure fair comparison. 3) *Action Steps* count the number of control steps the model takes to finish a task, such that a higher count can indicate added latency due to error detection, feedback, and resolution, or reflect unresolved failures where the task remains incomplete, and the action budget is exhausted.

1.5 Baselines, code, and material provided

Baselines. The baselines for our task include: 1) **RDT-1B** [32]: A diffusion-based VLA foundation model with 1.2 billion parameters, trained on a large-scale multi-robot corpus of over one million episodes across 46 datasets. It unifies heterogeneous robot action spaces within a shared latent representation for cross-robot learning. 2) π_0 [33]: A VLA model that combines a flow-matching architecture with a pre-trained vision-language model (VLM). The flow-matching component enables smooth trajectory learning between observations and actions, while the VLM provides strong visual-semantic grounding. 3) $\pi_{0.5}$ [34]: An VLA model that builds upon π_0 by incorporating partial fine-tuning of the pre-trained vision-language backbone within the flow-matching framework. This allows $\pi_{0.5}$ to adapt more effectively to downstream tasks while retaining strong generalization from pre-training. 4) **DreamZero** [9]: A WAM built on a video diffusion backbone that jointly models future world states and actions. By learning physical dynamics from diverse robot data, DreamZero achieves better generalization, supports real-time control, and enables cross-embodiment transfer. 5) **Motus** [7]: A unified latent action world model that integrates vision, language, and action understanding through a mixture-of-transformers architecture. By leveraging optical-flow-based latent actions and large-scale multimodal data, Motus achieves strong generalization and state-of-the-art performance across both simulated and real-world manipulation tasks.

Protocol of Releasing Code and Data We plan to release the starting kit through a dedicated GitHub repository that will contain all necessary materials for participants to easily join the competition. The repository will include: 1) Baseline code for model training and evaluation, 2) Data-loading and preprocessing tools for both simulated and real-world datasets, and 3) Sample simulated data together with a data-generation pipeline to enable participants to reproduce and extend training environments. In addition, we will release the real-world dataset and the complete simulated dataset (with data generator) on a public hosting platform (Hugging Face) to ensure accessibility and scalability.

1.6 Website, tutorial, and documentation

A dedicated competition website ([Guiliang:\(under construction\)](#)) will serve as the central hub for all information related to the event. The website will comprehensively present the competition overview, detailed timeline, participation steps, and relevant submission instructions. It will feature a FAQ/Tutorial section offering step-by-step guidance on installing the simulation environment, generating datasets, and testing models in both simulated and real-world settings. To ensure open communication, participants will be able to contact the organizers directly via robosyn-challenge@gmail.com, with responses and updates maintained regularly. All content, including documentation and tutorials, will be made available well in advance of the competition start date, and the website will go live within two weeks of acceptance notification. The official GitHub repository ([Guiliang:\(under construction\)](#)) will host the codebase, baseline models, and data tools, while an accompanying white paper will describe the competition design, problem formulation, and technical foundation supporting the challenge.

2 Organizational aspects

2.1 Protocol

Steps to Join the Competition. Participants begin by registering on the official competition website, which links to a dedicated GitHub repository containing the starting kit with baseline code, data-loading utilities, and a simulation environment. Teams can set up the environment locally or on cloud platforms using the provided Docker image or setup scripts, then access or generate both real and simulated datasets. They may develop and train models by extending or modifying provided baselines, validate performance in simulation, and ultimately deploy their models through a server-hosted HTTP inference endpoint connected to the competition system. During evaluation, sensory data from real robots are streamed to the participant’s endpoint, which returns action commands for real-world execution and benchmarking.

Competition Phases and Platform. The competition unfolds in three sequential phases. Phase I – Simulation Track allows participants to train and submit models within the simulation environment, generating public leaderboard results. Phase II – Real-World Validation invites the top-performing teams to submit inference endpoints for testing on standardized physical robotic systems under controlled conditions. Phase III – Final Evaluation and System Presentation challenges selected teams with unseen tasks, either on-site or remotely, to assess true generalization capabilities, with final results revealed during the closing session.

Cheating and Overfitting Prevention. To ensure fairness and scientific rigor, hidden test sets with varied backgrounds, lighting, and object configurations prevent data leakage and encourage robust generalization. All real-world evaluations are conducted server-side, under identical hardware and environmental conditions, and participants interact only through inference APIs without direct robot access. A single unified model is required across tasks, and intellectual property is protected since only model outputs (not internal code or weights) are transmitted. Submission frequency is limited to curb leaderboard overfitting. Collectively, these mechanisms ensure reproducible and trustworthy results across both simulated and real-world benchmarks.

2.2 Rules and Engagement

Contest Rules. The competition is open to all members of the NeurIPS community and the wider research public, welcoming individuals or teams from academia and industry worldwide. It consists of three phases: a Simulation Track for model development and public evaluation, a Real-World Evaluation for top teams to test models on standardized robotic setups, and a Final Evaluation involving hidden test environments with unseen tasks and real-robot trials.

Participants submit model checkpoints or host inference endpoints according to official API specifications, with limited daily submissions to prevent overfitting. Performance is assessed automatically using standardized metrics such as Success Rate, Inference Time, and Action Steps under consistent hardware and environmental conditions. Hidden test sets with novel variations ensure fairness and true generalization. All participants must adhere to the ethical guidelines and code of conduct.

Communication with Participants. Participants will engage with the organizers and community through openly accessible channels, including a GitHub repository for code releases and technical support, a public discussion forum (in the same GitHub link) for community interaction, and a designated email address for administrative inquiries (see Section 1.6). Official announcements, rule

clarifications, and schedule updates will be shared via the competition website, GitHub repository, email notifications, and pinned posts in the public forum. This integrated communication framework ensures transparent, inclusive, and equitable access to information and resources for all participants.

2.3 Schedule and readiness

Timeline competition. Our detailed timeline is presented as follows:

Preparation (May 15 – June 14, 2026): 1) Finalize simulation environments and generate full synthetic datasets. 2) Internal beta testing of the evaluation API and leaderboard.

Launch (June 15, 2026): 1) Official competition launch and website release. 2) Release of "Starting Kit," baseline code (π_0 , RDT-1B), and training data.

Development Phase (June 15 – Sept 15, 2026): 1) 90-day window for participants to develop and refine models. 2) Release of "Real-World Sample Set" for Sim2Real alignment. 3) Final submission deadline for model code and endpoints.

Evaluation & Analysis (Sept 16 – Oct 20, 2026): 1) Real-world validation on physical robots for top-performing teams. 2) Verification of reproducibility and final scoring.

Conclusion (Oct 21 – Oct 31, 2026): 1) Publication of final leaderboard and competition summary report. 2) Announcement of winners for NeurIPS 2026 presentation.

2.4 Competition promotion and incentives

RoboSynChallenge executes a multi-channel promotion strategy and a dedicated diversity initiative.

Promotion Plan. We will disseminate the Call for Participation through major academic mailing lists, including robotics-worldwide, ML-news, and the CVPR/ICCV lists. To foster engagement, we will promote the competition via posters and technical webinars showcasing our Starting Kit, as well as by presenting the challenge at leading 2026 robotics conferences such as ICRA and RSS. Our social media outreach will feature "Leaderboard Spotlights" on X (Twitter), LinkedIn, and WeChat, complemented by a featured Hugging Face blog post to engage the broader open-source research.

Incentives and Authorship. Participants will compete for a 20,000 USD prize pool and 5,000 USD in travel grants to attend NeurIPS 2026. The top three teams will receive a podium presentation slot at the NeurIPS workshop. Furthermore, we will lead a joint "findings paper" for a top-tier journal (e.g., JMLR or IJRR). Named authorship will be granted to the top five teams, while all other qualifying participants who outperform the baseline will be credited under a consortium author.

Diversity and Inclusion. To lower participation barriers, all evaluations will be conducted on our side, and a "Zero-Hardware" track will be provided, allowing researchers without access to physical robots to compete via simulation. We will offer cloud computing credits to teams from resource-constrained institutions and conduct targeted outreach to organizations such as Black in AI, LatinX in AI, and WiML. Importantly, 50% of the travel award budget will be reserved for participants from underrepresented groups and researchers from developing nations, ensuring diverse and inclusive representation at the conference.

3 Resources

Organizing team.

Provide a brief overview of all team members and their collective competencies within the organizing team. Please note that diversity in the organizing team is encouraged, so please elaborate on this aspect as well. Make sure to include: coordinators, data providers, platform administrators, baseline method providers, beta testers, and evaluators. Provide a short biography of all team members in appendix B.

Resources provided by organizers. To ensure the successful execution of the RoboSynChallenge, we have secured the following resources: 1) Computing: A cluster of 50 NVIDIA H100 GPUs, a cluster of 16 NVIDIA A800 GPUs, and 10 GeForce RTX 5090 servers for model training, dataset generation and leaderboard hosting for participant support. 2) Hardware: four identical dual-arm workstations featuring AgileX PiPER-X manipulators and RealSense D435i cameras (Figure 2) for physical validation and redundancy. Staff: A ten-person team including PhD researchers, a full-stack engineer, and lab technicians to manage the API, hardware resets, and weekly technical

"Office Hours." Sponsors: Chinese University of Hong Kong, Shenzhen Loop Area Institute (SLAI), DexForce, and industry partners providing a 20,000 prize pool and 5,000 in travel grants.

A Technical Details

A.1 Data Generation with EmbodiChain

EmbodiChain [25] is an end-to-end, GPU-accelerated, and modular platform for embodied AI research that integrates high-performance simulation, automated data pipelines, and flexible learning tools, thereby supporting rapid experimentation and effective Sim2Real transfer. It consists of three tightly coupled stages for scalable, diverse, and physically-grounded dataset generation.

1) *Generative Simulation of Learning Environment.* To overcome the limited diversity of manually designed simulation environments, EmbodiChain employs a two-stage generative framework. First, it synthesizes simulation-ready assets via generative models followed by multi-objective optimization to ensure geometric fidelity, physical plausibility, and simulator compatibility. Second, these assets are composed into fully functional scenes through gradient-based layout synthesis that optimizes object placement for physical realism and robot reachability. This process yields physically consistent, richly annotated environments that form the foundation of large-scale embodied data generation.

2) *Data Scaling via Domain Expansion.* Building on the generated environments, EmbodiChain scales embodied data by automatically generating and expanding robot interaction trajectories to improve coverage and robustness. It promotes functional diversity through reachability-aware sampling that selects kinematically feasible robot states maximizing task-space dissimilarity (e.g., in end-effector approach direction, contact geometry, and interaction outcomes), reducing trajectory homogenization common in teleoperation and conventional planners. To further strengthen robustness, a closed-loop error recovery module detects execution failures (e.g., slippage, misaligned grasps, boundary violations) and reactively replans corrective motions, which are relabeled and reintegrated as supervision for recovery behaviors.

3) *Sim2Real Generalization via Online Data Streaming.* To facilitate scalable Sim2Real transfer, EmbodiChain implements an *Online Data Streaming (ODS)* mechanism that continuously feeds diverse experiences from simulation into the learning loop. A streaming-based visual augmentation module perturbs lighting, textures, and sensor parameters on the fly, enriching perceptual diversity and mitigating overfitting to simulation-specific appearances. In parallel, an asynchronous shared-memory architecture enables real-time data exchange between simulation and learning processes through lock-free circular buffers, maximizing experience throughput and sample efficiency.

Table 3: Summary of simulated data randomization parameters.

Feature	Parameter	Description
Light	intensity_range: $[min, max]$	Lighting intensity variation.
	position_range: $[[x_{min}, y_{min}, z_{min}], [x_{max}, y_{max}, z_{max}]]$	Light source position randomization.
	color_range: $[[0.6, 0.6, 0.6], [1.0, 1.0, 1.0]]$	Light color variation.
Object location	init_pos: $[x, y, z]$	Object initial position.
	position_range: $[[-x, -y, -z], [+x, +y, +z]]$	Offset range from initial position.
Object orientation	rotation_range: $[min, max]$	Initial object rotation randomization.
Object material	base_color_range: $[[0.2, 0.2, 0.2], [1.0, 1.0, 1.0]]$	Object surface color variability.
Object size	scale: $[x, y, z]$	Uniform or axis-specific scaling.
Table	position_range: $[[0, 0, -0.04], [0, 0, 0.04]]$	Table height variation (± 4 cm).
	random_texture_prob: $p_{texture}$	Probability of random table texture.
	base_color_range: $[[r_{min}, g_{min}, b_{min}], [r_{max}, g_{max}, b_{max}]]$	Table color range.
Background color	base_color_range: $[[0.2, 0.2, 0.2], [1.0, 1.0, 1.0]]$	Background plate color variation.
Camera (intrinsic)	$f_{x, range}$	Focal length range along x-axis.
	$f_{y, range}$	Focal length range along y-axis.
Camera (extrinsic)	pos_range: $[[0, -0.02, 0], [0.02, 0.02, 0.01]]$	Random camera position (XYZ).
	euler_range: $[[-0.175, -0.175, -0.175], [0.175, 0.175, 0.175]]$	Camera rotation (roll-pitch-yaw).
Robot init.	eef_pos_range: $[[-0.01, -0.01, -0.01], [0.01, 0.01, 0]]$	End-effector position variation.
	$q_{pos, range}$: $([j_{1, min}, \dots], [j_{1, max}, \dots])$	Joint configuration randomization.
Distractors	Common items (e.g., bowls, cups, toys)	Distractors for scene complexity.

B Biography of all team members

Provide a short biography of all team members, stressing their competence for their assignments in the competition organization. The section does not count towards the eight pages limit.

Table 4: Summary of Task Descriptions, Steps, and Success Criteria

Task Description	Steps	Success Criteria
Click Bell	(1) Move the arm toward the bell. (2) Align and press the button. (3) Return the arm to the initial position.	The button has been successfully pressed within a limited number of action steps.
Items Hand-Over and Place	(1) Grasp the pen. (2) Pass the pen to another arm. (3) Place the pen in the brush pot.	The pen has been successfully placed in the brush pot within a limited number of action steps.
Dual-Arm Water Pouring	(1) Grasp the water bottle and the cup. (2) Pour the water from the water bottle into the cup. (3) Place both objects securely on the table.	The water has been successfully poured into the cup, and both items are placed on the table within a limited number of action steps.
Table Rearrangement	(1) Grasp the spoon and the fork. (2) Arrange them neatly on both sides of the plate	The spoons and forks have been arranged around the plate within a limited number of action steps.
Basket Pick-and-Place	(1) Grasp the milk box. (2) Place the milk box in the basket. (3) Grasp the basket. (4) Place the basket in the center of the table.	The milk box has been placed into the basket, and the basket has been set at the table center within a limited number of action steps.
Drawer Open and Place	(1) Grasp the handle. (2) Open the drawer. (3) Grasp an item. (4) Place the item in the drawer. (5) Close the drawer.	The item has been successfully placed in the drawer within a limited number of action steps.
Mixer Operating	(1) Grasp the beaker. (2) Place the beaker on the mixer. (3) Start the mixer.	The beaker was placed on the mixer and stirring successfully started within a limited number of action steps.
Item Assembly	(1) Grasp the silicone tubes. (2) Adjust them to a horizontal position. (3) Move one tube forward. (4) Move the other tube and splice together.	The two silicone tubes have been successfully joined together within a limited number of action steps.
Manipulate Pipette	(1) Grasp the pipette. (2) Insert pipette into the first beaker. (3) Another arm presses the pipette button. (4) Move pipette to another beaker. (5) Press the pipette button again.	Liquid has been successfully transferred from one beaker to another within a limited number of action steps.
Sample Loading	(1) Grasp the test tube. (2) Pass the test tube to another arm. (3) Place the test tube into the rack.	The test tube has been successfully placed in the rack within a limited number of action steps.

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